The documentation and process conversion measures necessary to comply with this revision shall be completed by 19 October 1999.

INCH-POUND

MIL-PRF-19500/240G
19 July 1999
SUPERSEDING
MIL-S-19500/240F
21 February 1994

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, RECTIFIER,
TYPES 1N645-1, 1N647-1, 1N649-1, 1N645UR-1, 1N647UR-1, 1N649UR-1
JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for general purpose silicon diode. Three levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See 3.3 and figure 1 (similar to DO-41 for types 1N645-1, 1N647-1 and 1N649-1) and figure 2 (similar to DO-213AA for types 1N645UR-1, 1N647UR-1, and 1N649UR-1).

1.3 Maximum ratings.

Type 1/	V_{RSM}	V_{RWM}	I_O T_A = +150°C	I_O 2/ T_A = +25°C	I_{FSM} t_p = 1/120 s T_A = +25°C	T_J	T_{STG}	Barometric pressure, reduced
	V (pk)	V (pk)	mA	mA	A	°C	°C/W	mm Hg
1N645-1	270	225	150	400	5	-65 to +175	-65 to +175	8
1N647-1	480	400	150	400	5	-65 to +175	-65 to +175	8
1N649-1	720	600	150	400	5	-65 to +175	-65 to +175	8

1/ Electrical characteristics for UR suffix devices are identical to the corresponding non-UR suffix devices unless otherwise specified..

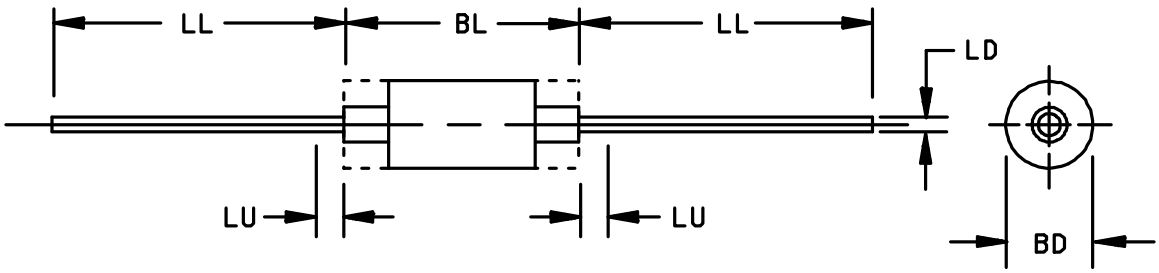
2/ Derate 2.0 mA/°C between 25°C to 150°C. Derate 6 mA/°C between 150°C to 175°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAC, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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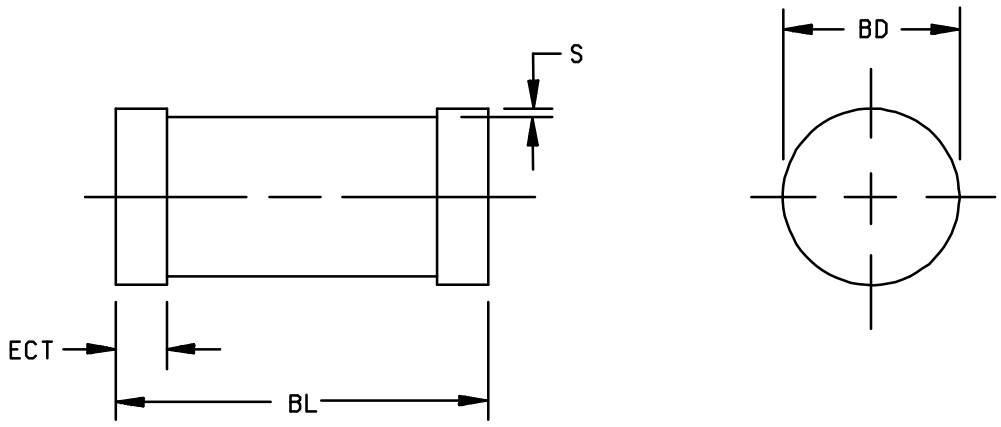


Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
BD	0.055	0.090	1.40	2.29	5
BL	0.120	0.200	3.05	5.08	
LD	0.018	0.022	0.46	0.56	
LL	1.000	1.500	25.40	38.10	
LU		0.050		1.27	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Symbol BD shall be measured at the largest diameter.
4. Within LU lead diameter may vary to allow for flash, lead finish build-up, and minor irregularities other than heat slugs.
5. The specified lead diameters apply in the zone between .050 (1.27 mm) from the diode body to the end of the lead. Outside of this zone the lead diameter shall not exceed the maximum dimension BD.
6. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions for types 1N645-1, 1N647-1, and 1N649-1.

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.130	.146	3.30	3.70
BD	.063	.067	1.60	1.70
ECT	.016	.022	0.41	0.55
S	.001 min		0.03 min	

NOTES:
1. Dimensions are in inches.
2. Metric equivalent are given for general information only.
3. Dimensions are presolder dip.
4. Referencing dimension S, minimum clearance of glass body to mounting surface on all orientations.
5. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 2. Physical dimensions for types 1N645UR-1, 1N647UR-1, and 1N649UR-1.

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1.4 Primary electrical characteristics at T_A = +25°C, unless otherwise specified.

Type 1/	V_F at I_F = 400 mA dc, 2% duty cycle, 8.3 ms max pulse width	I_R at T_A = 25°C		I_R at T_A = 150°C	
	V dc (max)	µA dc (max) at V_R		µA dc (max) at V_R	
1N645-1	1.0	.050	225 V dc	25	225 V dc
1N647-1	1.0	.050	400 V dc	25	400 V dc
1N649-1	1.0	.050	600 V dc	25	600 V dc

1/ Electrical characteristics for UR suffix devices are identical to the corresponding non-UR suffix devices unless otherwise specified..

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

DEPARTMENT OF DEFENSE

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Automated Printing Service, 700 Robbins Avenue, Building 4D (DPM-DODSSP), Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-PRF-19500.

3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500 and figures 1, 2 herein.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-PRF-19500, MIL-STD-750, and herein.

3.2.2 Dash one construction. Dash one (-1) devices shall be metallurgically bonded double plug construction in accordance with the requirements of category I, II, or III (see MIL-PRF-19500, paragraph 30.14 and subparagraphs).

3.4 Marking. Devices shall be marked in accordance with MIL-PRF-19500. At the option of the manufacturer, the marking of the country of origin may be omitted from the body of the diode.

3.4.1 Polarity. The polarity shall be indicated with a constrasting color band to denote the cathode end. No color coding will be permitted.

3.4.2 Marking of UR version devices. For UR version devices only, all marking (except polarity) may be omitted from the body, but shall be retained on the initial container.

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.6 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I.

3.7 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.4).

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

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4.3 Screening (JANTXV, JANTX levels only). Screening shall be in accordance with MIL-PRF-19500 (table IV) and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement JANTX and JANTXV levels
3c	Thermal impedance (see 4.3.2)
9	Not applicable
11	I_{R1}, V_{F1}
12	See 4.3.1, t = 48 hours
13 1/	Subgroup 2 of table I herein, $\Delta I_{R1} \leq$ 100 percent of initial reading or 10 nA dc, whichever is greater. $\Delta V_F \leq \pm$0.025 V dc change from initial reading

1/ Thermal impedance need not be repeated at this step.

4.3.1 Power burn-in conditions. All devices shall be operated under one of the following conditions:

Type 1/	MIL-STD-750, method 1038 condition B (a), T_A = 125°C, f = 60 Hz, (see 4.5.2)	MIL-STD-750, method 1038 condition B (a), T_A = 25°C, f = 60 Hz, (see 4.5.2)	MIL-STD-750, method 1038 condition B (b), T_A = 125°C
1N645-1	V_R = 225 V(pk), I_O = 200 mA	V_R = 225 V(pk), I_O = 400 mA	I_F = 200 mA
1N647-1	V_R = 400 V(pk), I_O = 200 mA	V_R = 400 V(pk), I_O = 400 mA	I_F = 200 mA
1N649-1	V_R = 600 V(pk), I_O = 200 mA	V_R = 600 V(pk), I_O = 400 mA	I_F = 200 mA

1/ Electrical characteristics for UR suffix devices are identical to the corresponding non-UR suffix devices unless otherwise specified.

4.3.2 Thermal impedance $Z_{\theta JX}$ measurements for screening. Thermal impedance ZθJX measurements shall be performed in accordance with MIL-STD-750, method 3101. The maximum limit shall not exceed the group A, subgroup 2 limit for $Z_{\theta JX}$ in screening (table II MIL-PRF-19500) and atypical devices shall be removed.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF19500 and as specified herein. Group A inspection shall be performed on each sublot.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein.

The following test conditions shall be used for $Z_{\theta JX}$, group A inspection:

a. I_M measure current ..... 1 mA - 10 mA.

b. I_H forward heating current ..... 0.5 A - 1.0 A.

c. t_H heating time..... 10 ms.

d. t_{MS} measurement delay time..... 100 µs. (maximum

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 and paragraphs 4.4.2.1 herein. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 herein. Delta measurements shall be in accordance with table II herein.

4.4.2.1 Group B inspection, table VIb (JAN, JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1027	I_O = 40 mA min, f = 50 – 60 Hz, T_A = 150°C (see 4.5.2). V_R = 225 V(pk) for 1N645-1; V_R = 400 V(pk) for 1N647-1; V_R = 600 V(pk) for 1N649-1
	4066	Not required
B5	3101 or 4081	L = .375", $R_{\theta JL}$ = 250°C/W; L = 0", $R_{\theta JC}$ = 100°C/W (surface mount); No moving air permitted.
B6	1032	T_A = 175°C

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, group A, subgroup 2 herein. Delta measurements shall be in accordance with table II herein.

Subgroup	Method	Condition
C2	2036	Tension: Test condition A; weight = 2 pounds, t = 30 ±3 s. Lead fatigue: Test condition E. (Terminal strength not applicable to surface mount devices.)
C6	1026	I_O = 150 mA, f = 50 – 60 Hz, T_A = 150°C (see 4.5.2). V_R = 225 V(pk) for 1N645-1; V_R = 400 V(pk) for 1N647-1; V_R = 600 V(pk) for 1N649-1

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4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IX of MIL-PRF-19500 and as follows. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

Subgroup	Method	Condition	Sampling plan
E1, E2, E3	---	Not applicable	
E4	3101 or 4081	L = .375", $R_{\theta JL}$ = 250°C/W for non surface mount; L = 0", $R_{\theta JEC}$ = 100°C/W for surface mount. No moving air permitted.	10 devices, c = 0
E5	1001		10 devices, c = 0

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Steady-state operation life. A half-sine wave of the specified peak voltage shall be impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall not be greater than 180 degrees nor less than 150 degrees.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Thermal impedance	3101	See 4.4.1	$Z_{\theta JX}$		35	°C /W
Forward voltage	4011	I_F = 400 mA dc, (Pulsed, see 4.5.1), 2 percent duty cycle maximum, 8.3 ms maximum pulse width	V_{F1}	0.8	1.0	V dc
Reverse current	4016	DC method	I_{R1}			
1N645-1		V_R = 225 V dc			0.05	µA dc
1N647-1		V_R = 400 V dc			0.05	µA dc
1N649-1		V_R = 600 V dc			0.05	µA dc
Subgroup 3						
High temperature operation		T_A = 150°C				
Forward voltage	4011	I_F = 400 mA dc, (Pulsed, see 4.5.1), 2 percent duty cycle maximum, 8.3 ms maximum pulse width	V_{F2}	0.7	0.95	V dc
Reverse current	4016	DC method	I_{R2}			
1N645-1		V_R = 225 V dc			25	µA dc
1N647-1		V_R = 400 V dc			25	µA dc
1N649-1		V_R = 600 V dc			25	µA dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 – Continued						
Low temperature operation		T_A = -55°C				
Forward voltage	4011	I_F = 400 mA dc, (Pulsed, see 4.5.1), 2 percent duty cycle maximum, 8.3 ms maximum pulse width	V_{F3}		1.2	V dc
Subgroup 4						
Reverse current at peak reverse voltage	4016	AC method, f= 60 Hz	I_{RM1}			
1N645-1		V_R = 225 V(pk)			50	µA(pk)
1N647-1		V_R = 480 V(pk)			50	µA(pk)
1N649-1		V_R = 720 V(pk)			50	µA(pk)
Capacitance	4001	V_R = 4 V dc, f = 1 MHz, V_{sig} = 50 mV(pk-pk) maximum	C		20	pF
Subgroup 5						
Not applicable						
Subgroup 6						
Surge current	4066	Mounting conditions in accordance with MIL-STD-750, test method 1026 T_A = 25°C, I_{FSM} = 5 A, I_O = 150 mA, ten 1/120 s surges, 1 surge per minute.				
Electrical measurements		In accordance with Group A, subgroup 2 herein				

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 7						
Reverse current, average	4046	T_A = 150°C, I_O = 150 mA	$I_{R (AV)}$			
1N645-1		V_R = 225 V(pk)			100	µA
1N647-1		V_R = 480 V(pk)			100	µA
1N649-1		V_R = 720 V(pk)			100	µA

1/ Electrical characteristics for UR suffix devices are identical to the corresponding non-UR suffix devices unless otherwise specified.

TABLE II. Groups A, B, and C delta measurements. 2/ 3/ 4/

Step	Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Reverse current	4016	DC method;	ΔI_{R1} 2/			
	1N645-1		V_R = 225 V(pk)		100 % of initial value or ±.010 µA dc, whichever is greater		
	1N647-1		V_R = 400 V(pk)				
	1N649-1		V_R = 600 V(pk)				

1/ Electrical characteristics for UR suffix devices are identical to the corresponding non-UR suffix devices unless otherwise specified.

2/ Devices exceeding group A limits for this test shall not be acceptable.

3/ The delta measurements for table VIb (JAN, JANTX, and JANTXV) of MIL-PRF-19500 are as follows:

a. Subgroups 3 and 6, see table II herein, step 1.

4/ The delta measurements for table VII of MIL-PRF-19500 are as follows:

a. Subgroup 6, see table II herein, step 1 (for JAN, JANTX, and JANTXV).

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.2.1).

b. Lead finish (see 3.3.1).

c. Type designation and product assurance level.

d. Packaging requirements (see 5.1).

6.2.1 Substitutability. Non dash-one devices have been deleted from this specification. Dash-one devices are a direct substitute for non dash-one devices and are preferred. Devices in stock are acceptable provided the date code does not exceed 10 February 1993. Existing supplies of the non dash-one parts may be used until exhausted.

6.2.2 Cross reference substitution list. JANS level will no longer be built to MIL-PRF-19500/240. Devices in stock are acceptable provided the date code does not exceed the date of implementation of MIL-PRF-19500/240F. Devices required for space flight applications are found in MIL-PRF-19500/587. Existing supplies of 1N645-1, 1N647-1, and 1N649-1 can be used until existing supplies are exhausted. A PIN for PIN replacement table follows, and these devices are directly interchangeable:

Non-preferred PIN	Preferred PIN
JANS1N645-1	JANS1N6661
JANS1N647-1	JANS1N6662
JANS1N649-1	JANS1N6663

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.

Custodians:
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Preparing activity:
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/240G	2. DOCUMENT DATE *(YYYYMMDD)*

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, DIODE, SILICON, RECTIFIER, TYPES 1N645-1, 1N647-1, 1N649-1, 1N645-1UR, 1N647-1UR, 1N649-1UR, JAN, JANTX, AND JANTXV

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) DSN *(If applicable)*	7. DATE SUBMITTED *(YYYYMMDD)*

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE *(Include Area Code* (1) Commercial 614-692-0510	(2) DSN 850-0510
c. ADDRESS *(Include Zip Code)* DSCC-VAC 3990 East Broad Street Columbus, Ohio 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703)767-6888 DSN 427-6888	

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